MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 01** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal choice.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If the circumference and the area of a circle are numerically equal, then the radius of the circle is: [1]

(A) 1 unit (B) 2 units (C) π units (D) 4 units

Q2. The area of the circle that can be inscribed in a square of side 6 cm is: [1]

(A) 36π cm^2 (B) 18π cm^2 (C) 12π cm^2 (D) 9π cm^2

Q3. If the area of a circle is equal to the sum of the areas of two circles of diameters 10 cm and 24 cm, then the diameter of the larger circle is: [1]

(A) 34 cm (B) 26 cm (C) 17 cm (D) 14 cm

Q4. The perimeter of a semicircular protractor of radius r is given by: [1]

(A) πr (B) $\pi r + r$ (C) $\pi r + 2r$ (D) $2\pi r + 2r$

Q5. **Assertion (A):** If the areas of two circles are in the ratio 9 : 16, then their circumferences are in the ratio 3 : 4.
Reason (R): If the areas of two circles are A_1 and A_2, then the ratio of their circumferences is $\sqrt{A_1} : \sqrt{A_2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. The area of a circular playfield is 616 m^2. Find its circumference. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find: (i) the length of the arc, (ii) the area of the sector formed by the arc, and (iii) the area of the segment formed by the corresponding chord. $\left(\text{Use } \pi = \frac{22}{7}, \sqrt{3} = 1.732\right)$ [3]

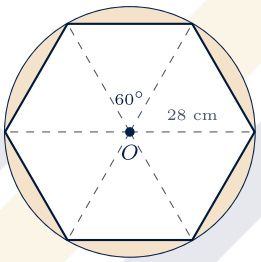
Q8. A chord of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14, \sqrt{3} = 1.73$) [3]

Q9. A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding minor segment of the circle. (Use $\pi = 3.14, \sqrt{3} = 1.73$) [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Round Dining Table Cover with Hexagonal Border Designs

A craftsman creates an embroidered circular table cover for a round dining table. The cover of radius $r = 28$ cm has six identical floral border designs along its perimeter, leaving a central regular hexagonal region.



(i) Find the central angle subtended by each of the six design segments at the centre of the table cover. [1]

(ii) Calculate the area of the equilateral triangle formed by connecting the centre to the endpoints of one design. [1]

(iii) Calculate the total area of all six embroidered designs, OR find the total cost of embroidering the designs at the rate of 0.50 per cm^2. $\left(\text{Use } \sqrt{3} = 1.732, \pi = \frac{22}{7}\right)$ [2]

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